

Algebra II with Trigonometry

Chapter 6.4

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4533650_alg-2trig-h-chp-6-4-note.pdf](#)

Here are 10 mental math questions based on the PDF, testing understanding of trigonometric graphs, periods, vertical asymptotes, and transformations:

1. Period of Tan and Cot

What is the period of the functions $y = \tan(x)$ and $y = \cot(x)$?

2. Period of Sec and Csc

What is the period of the functions $y = \sec(x)$ and $y = \csc(x)$?

3. Location of Vertical Asymptotes for Tan(x)

At what x -values within one period does $y = \tan(x)$ have vertical asymptotes?

4. Location of Vertical Asymptotes for Cot(x)

At what x -values within one period does $y = \cot(x)$ have vertical asymptotes?

5. Secant Asymptote Location

What is the form of the vertical asymptotes for $y = \sec(x)$?

6. Cosecant Asymptote Location

$y = \csc(x)$ is undefined at which x -values?

7. New Period of $y = \tan(3x)$

What is the period of $y = \tan(3x)$?

8. New Period of $y = \sec(2x)$

What is the period of $y = \sec(2x)$?

9. One-Period Interval for $y = \cot(kx)$

For $y = \cot(kx)$, what is the correct length of one period?

10. Transformation: Shifted Tangent

For $y = \tan(2x - \pi)$, what interval should be used to graph one period?

Let me know if you would like answers or further explanations!